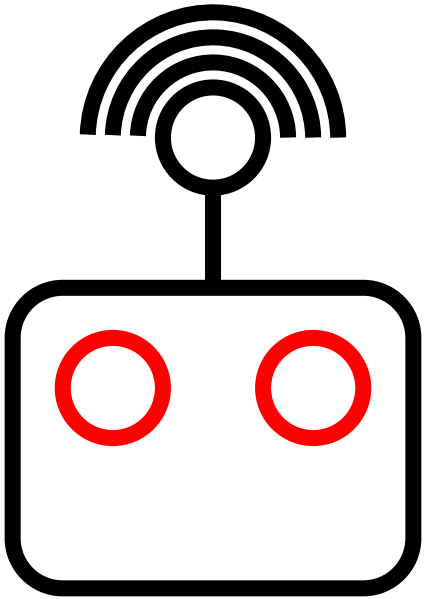


WaCS-1 ESP32 Remote Robotics Control System

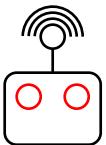
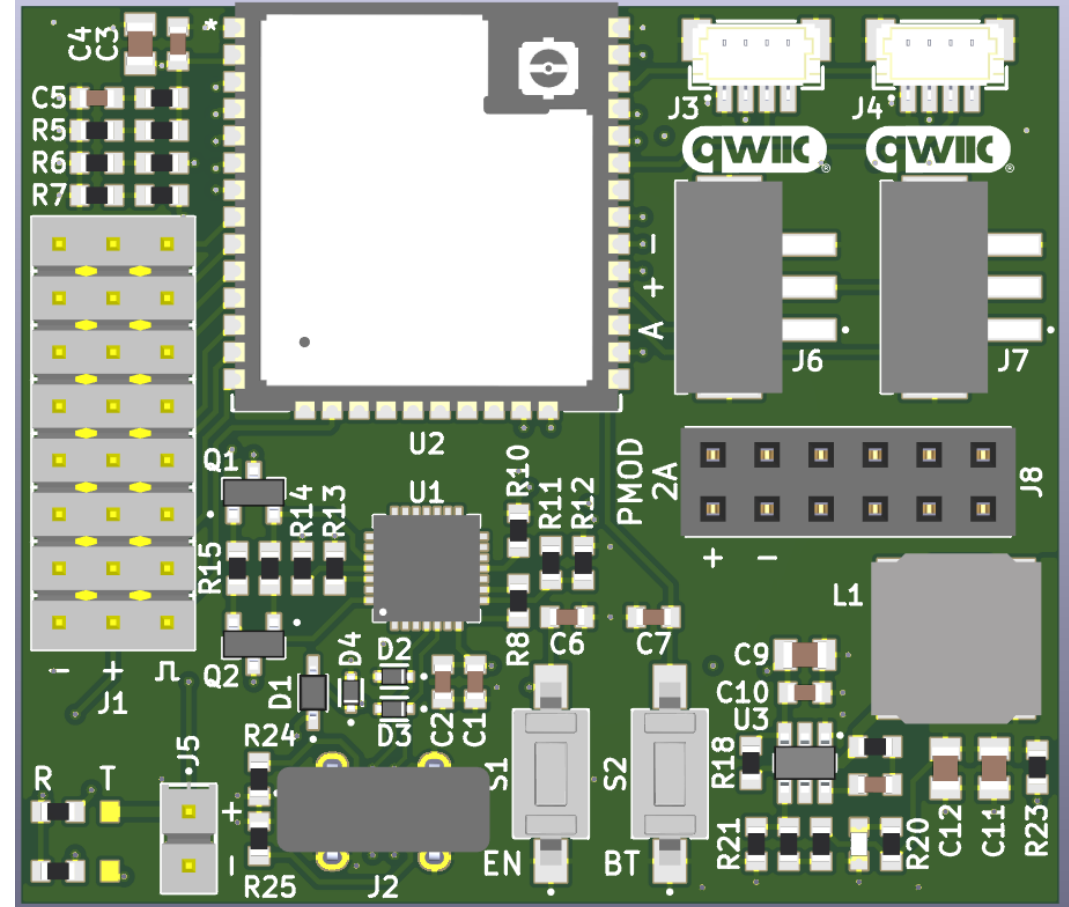
Dan Tatum

27 February 2025



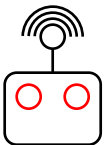
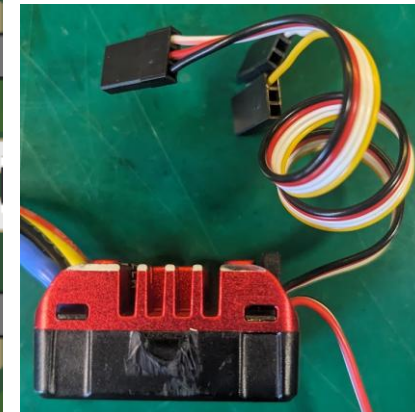
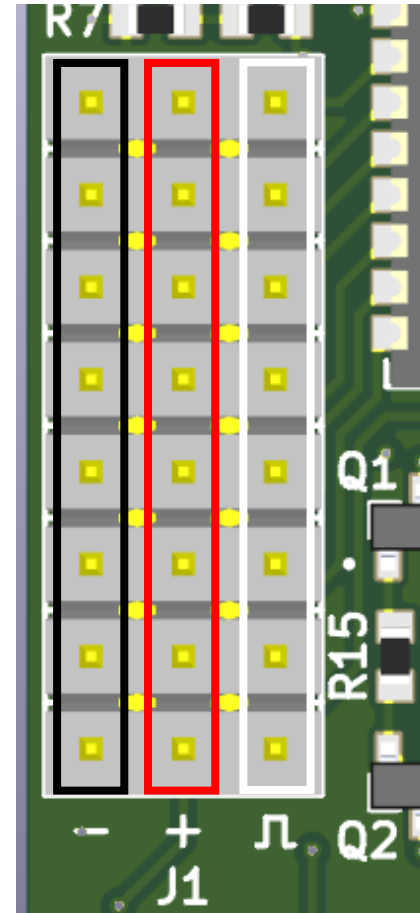
WaCS-1 ESP32 Remote Robotics Control System Overview

- Up to 8 PWM servos
- Dual SparkFun Qwiic / Adafruit STEMMA I²C connections
- Dual Adafruit STEMMA analog connections
- Digilent PMOD 2A SPI connection
- USB-C connector for firmware upload
- U.FL connector for improved antenna performance
- Voltage monitors for on board 3.3V supply, ESC, and external battery
- High efficiency switch mode power supply
- Reconfigurable to support potentiometer inputs from remote controller
- 1.92" x 1.66" PCBA area



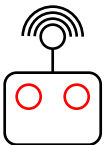
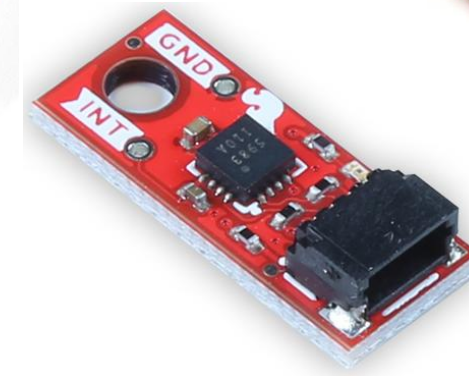
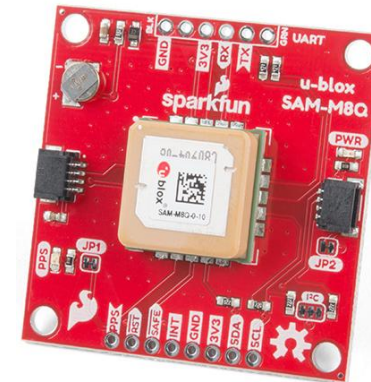
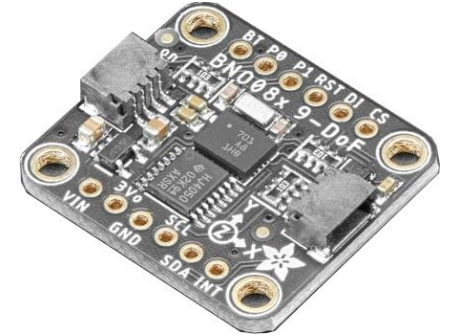
Up to 8 PWM Servos

- Supports up to 8 PWM based servos
- 0.1" square pin headers
- ESC power pass through to servos
- Controller powered by ESC



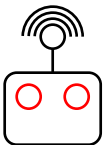
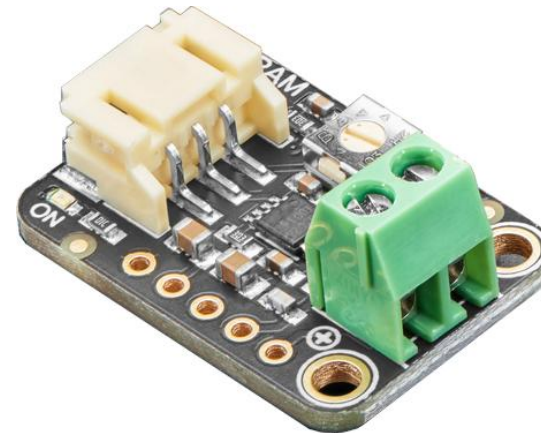
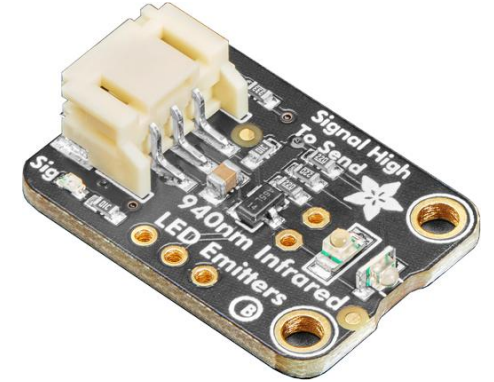
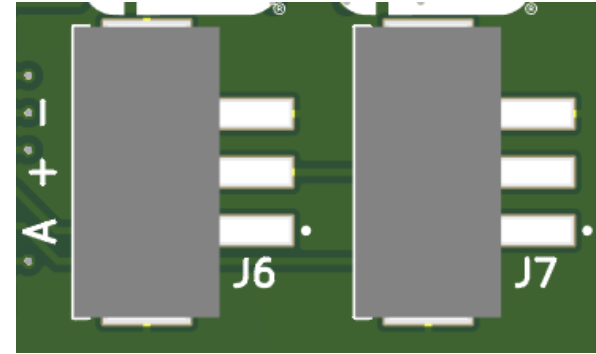
Dual SparkFun QWIIC / Adafruit STEMMA QT I²C Connections

- A world of possibilities with readily available off-the-shelf components
- JST type SH 4-pin 1mm connector
- GPS, IMU, screens, sensors, etc.
- Developers can focus on software and ditch the rat's nest
- <https://www.sparkfun.com/categories/399>
- <https://www.adafruit.com/category/1018>



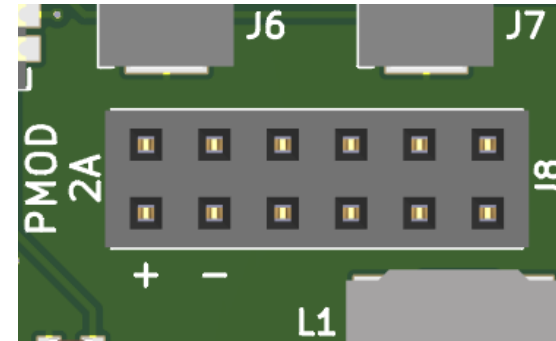
Dual Adafruit STEMMA Analog Connections

- Analog and GPIO
- A world of possibilities with readily available off-the-shelf components
- JST type PH 3-pin 2mm connector
- IR communication, buttons, amplifiers, etc.
- <https://www.adafruit.com/category/1019>

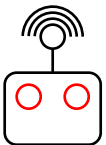
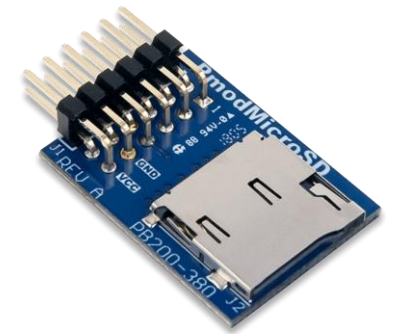
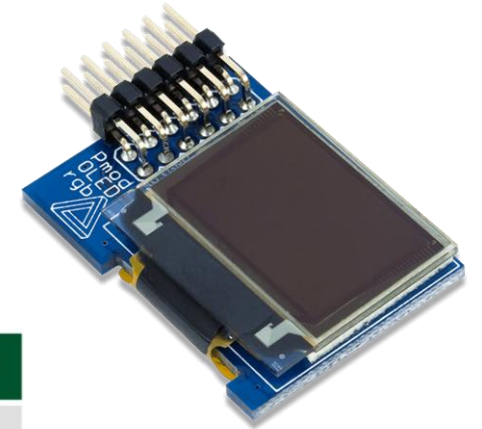


Digilent PMOD 2A SPI Connection

- Dual SPI connections
- A world of possibilities with readily available off-the-shelf components
- 2x6 pin 0.1" square receptacle
- <https://digilent.com/shop/products/fpga-boards/expansion-modules/pmods/>

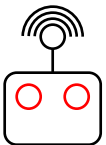
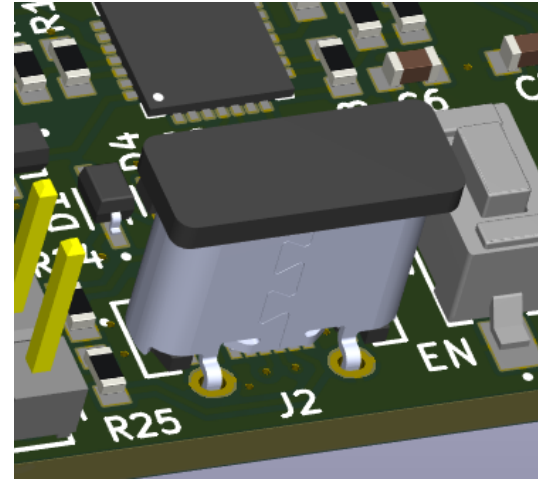


Pin #	Signal	Direction	Alternate Signal	Direction
1	CS	Out	-	-
2	MOSI	Out	-	-
3	MISO	In	-	-
4	SCK	Out	-	-
5	GND			
6	VCC			
7	GPIO	In/Out	INT	In
8	GPIO	In/Out	RESET	Out
9	GPIO	In/Out	CS2	Out
10	GPIO	In/Out	CS3	Out
11	GND			
12	VCC			



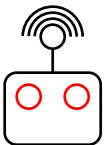
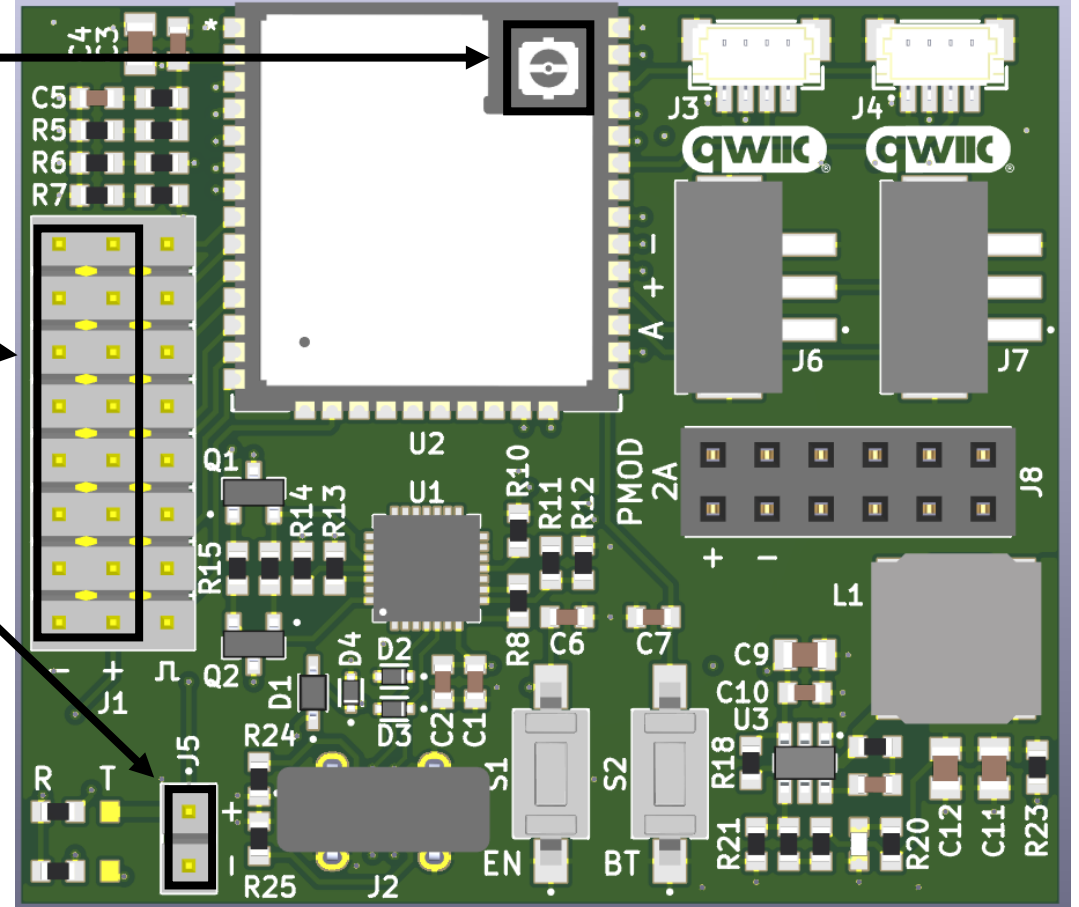
USB-C connector for firmware upload

- Common, current, reversible connection
- Integrated Silicon Labs CP2102 USB to serial converter
- No hunting for old, oddball cables or adapters
- Mounted perpendicular to board for easy in-unit access
- USB 2.0 for easy compatibility



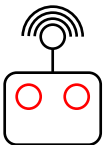
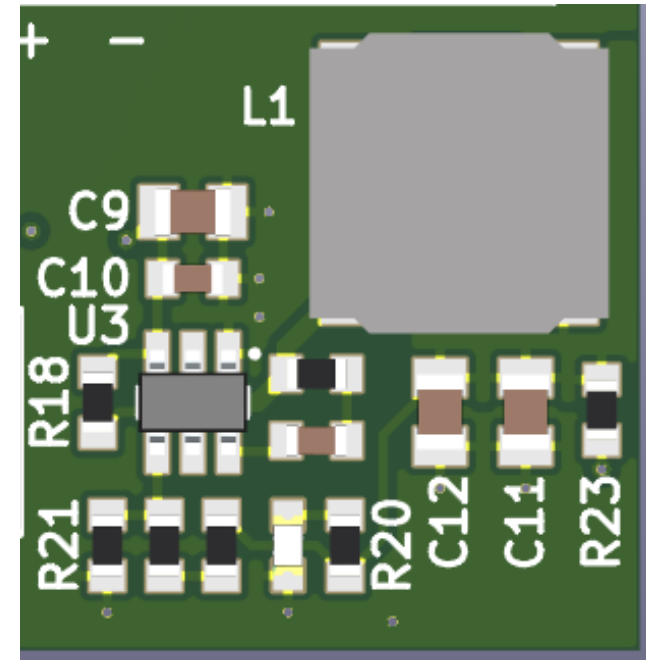
Additional Connections

- U.FL connector for improved antenna performance
- Voltage monitors for on board 3.3V supply, ESC, and external battery



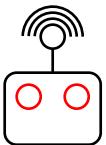
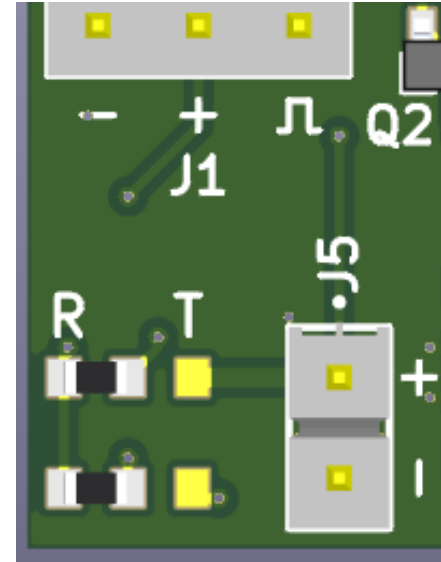
High efficiency switch mode power supply

- Operational input voltage 4.2V-18V
- 3.3V, 2A output to PCBA and external accessories

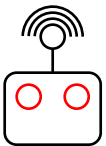


Reconfigurable for Potentiometer Inputs from Remote Controller

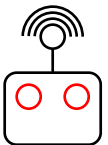
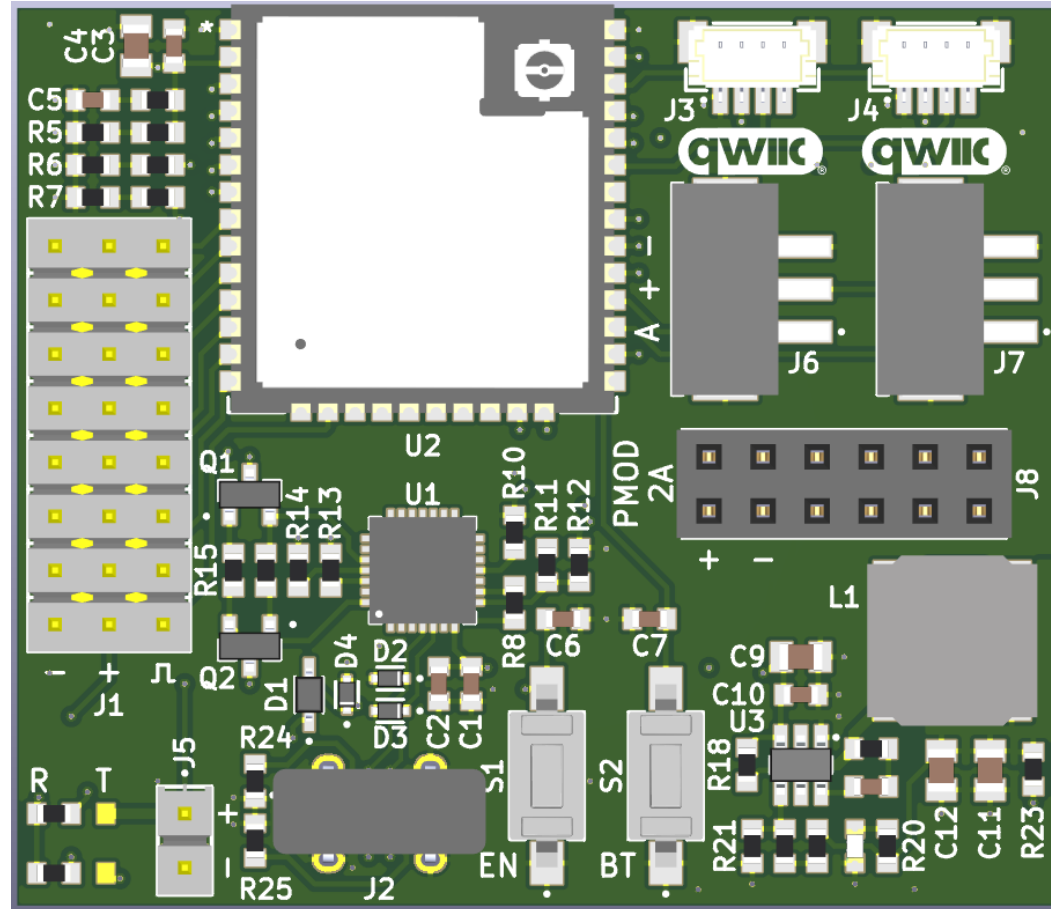
- Populate R1, R22 to left side pads
- Input power sourced from J5 external power monitoring input
- J1 sources 3.3V to external potentiometers and connects wiper to ADC input
- Inputs reconfigured in software



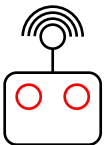
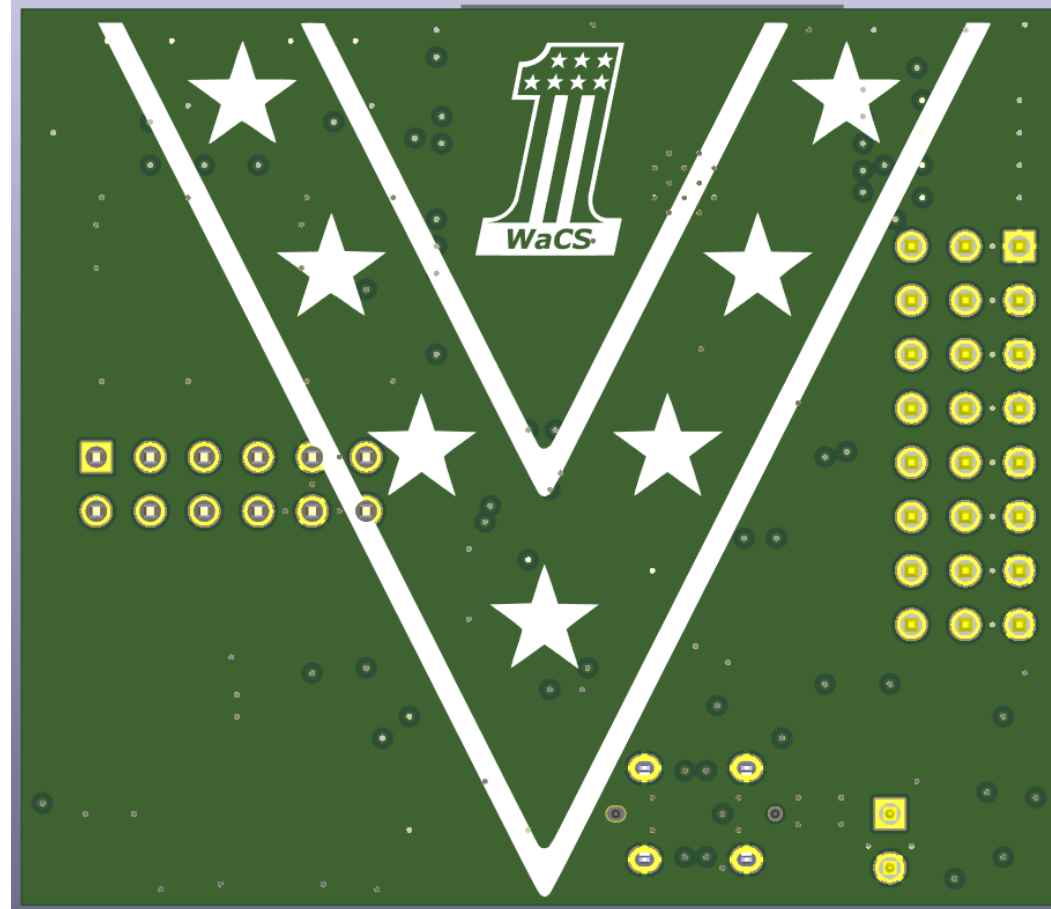
PCBA 3D View



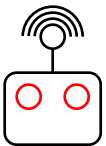
PCBA Front

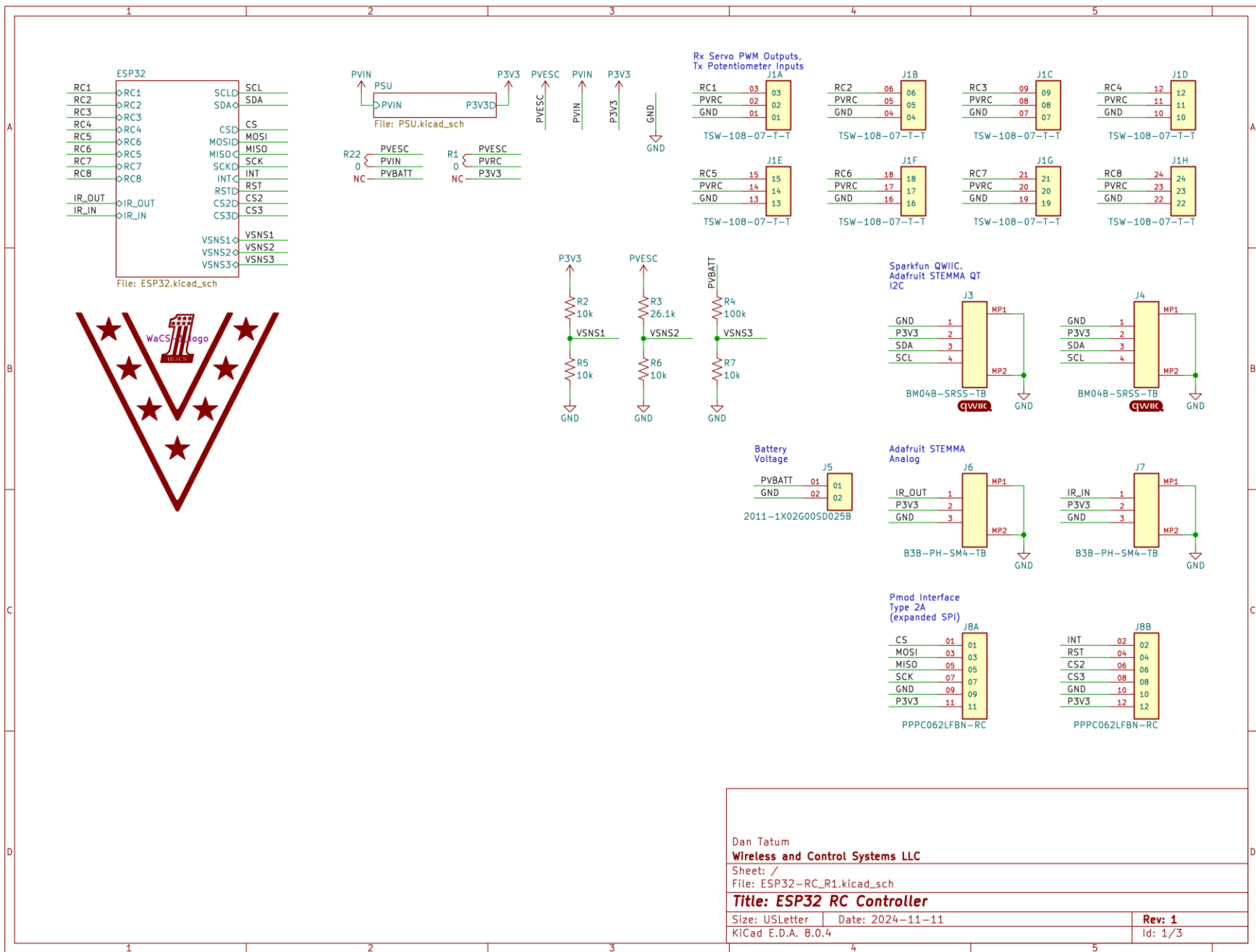


PCBA Back



PCBA Schematic





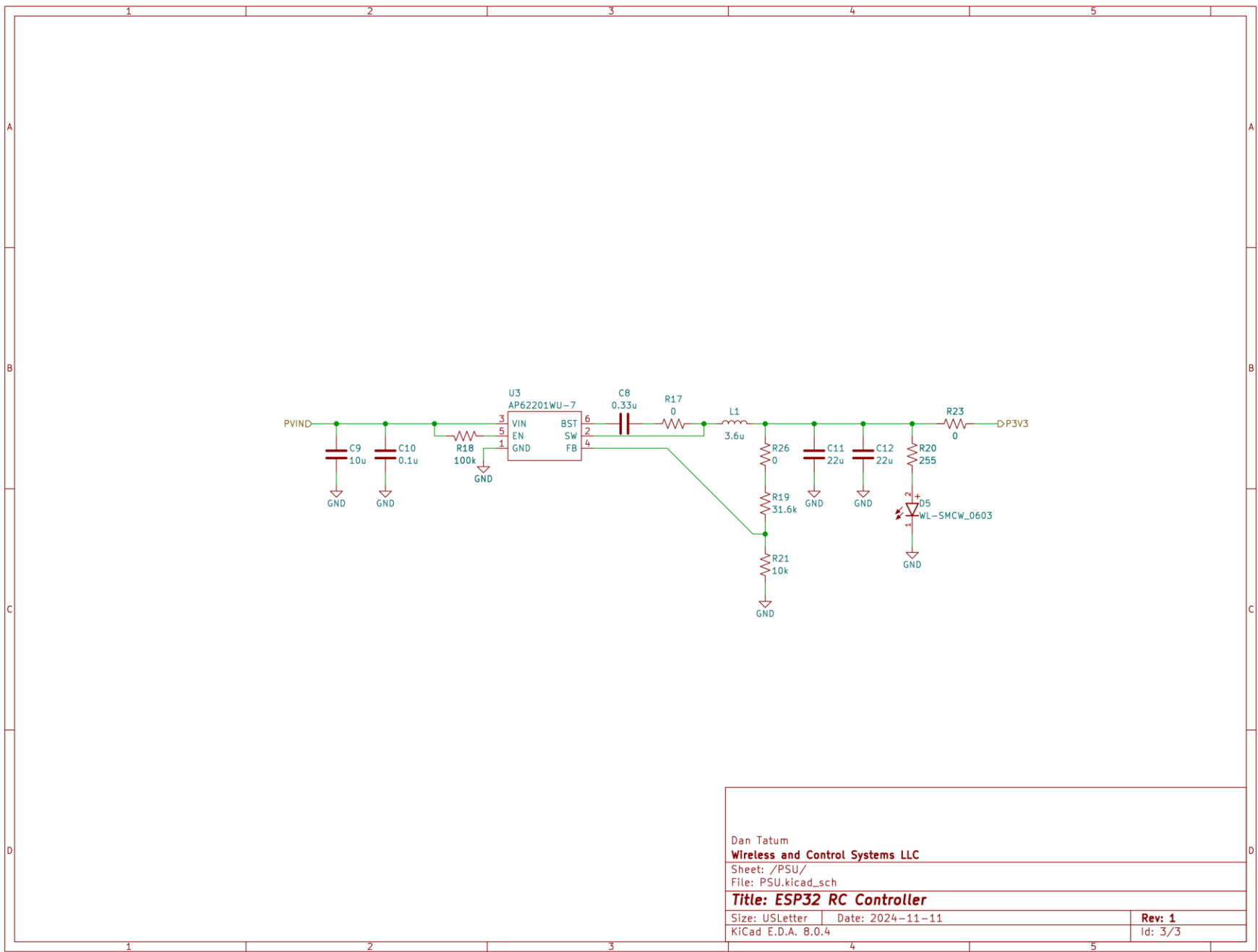
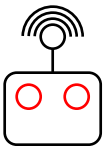
Dan Tatum
Wireless and Control Systems LLC

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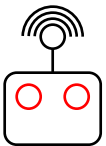
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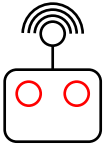
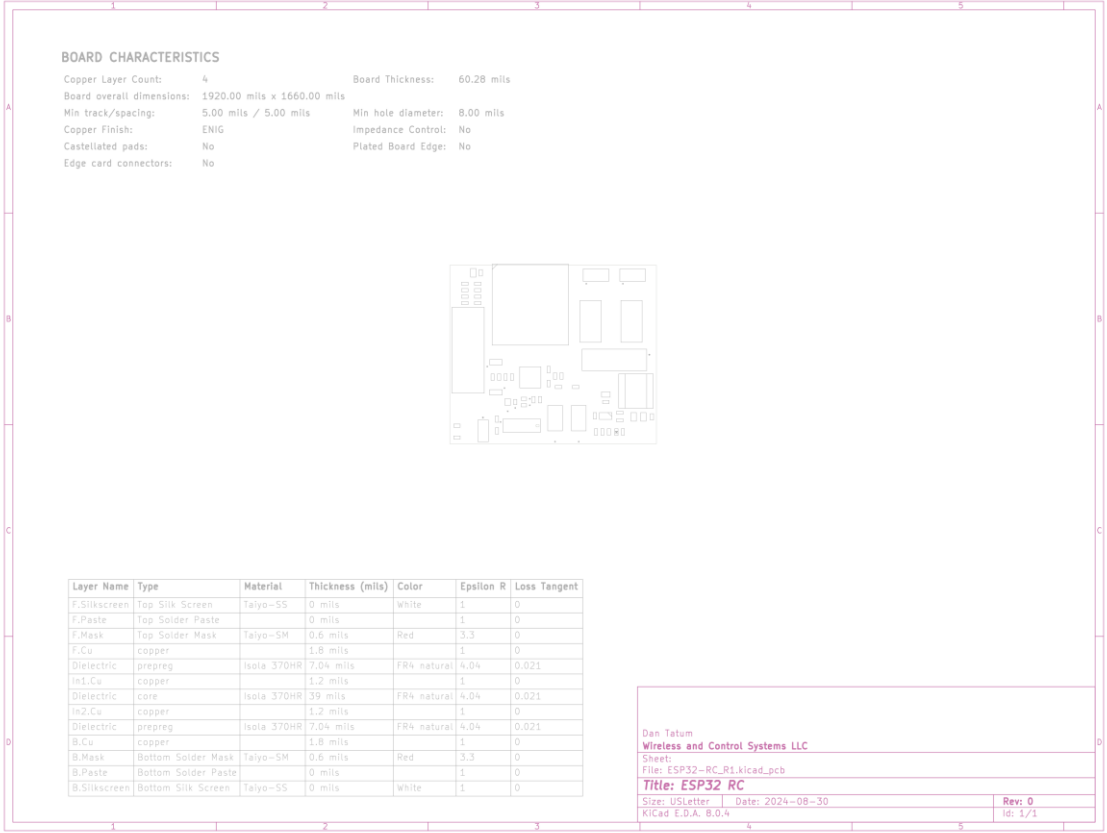
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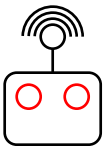
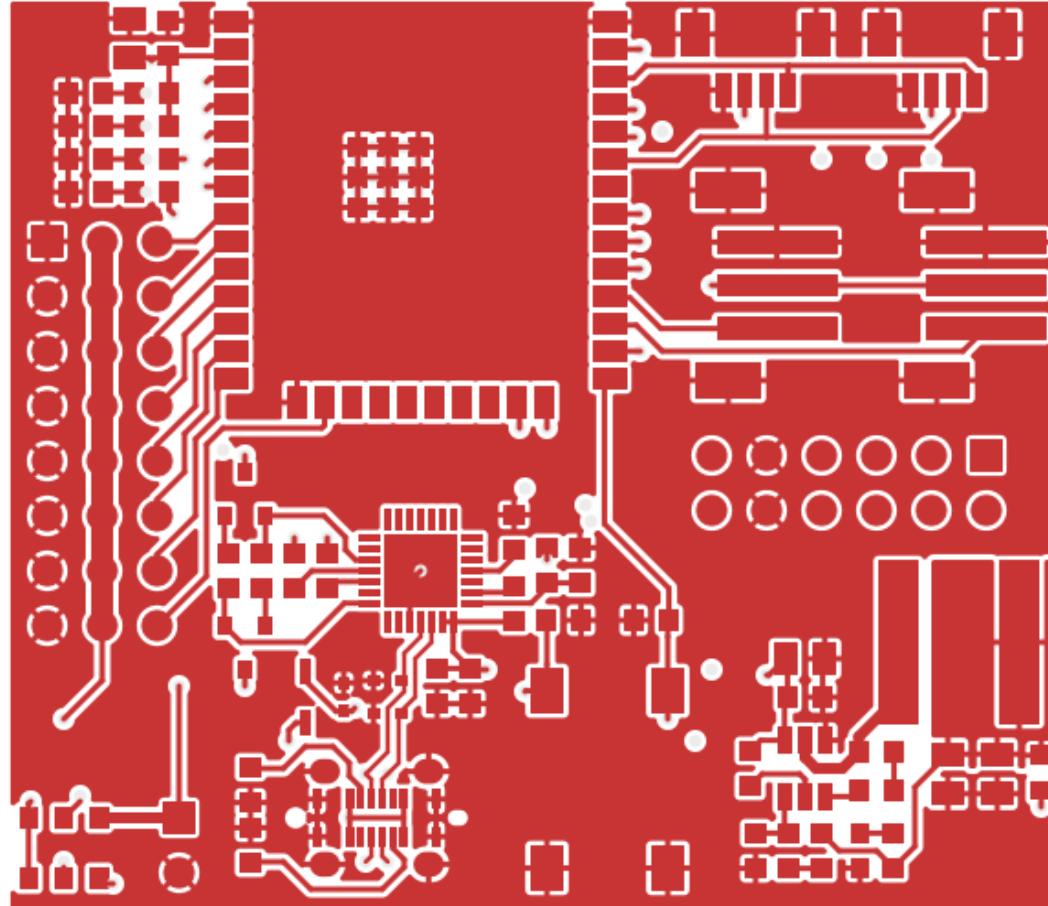
PCBA Gerbers



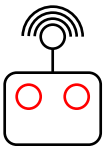
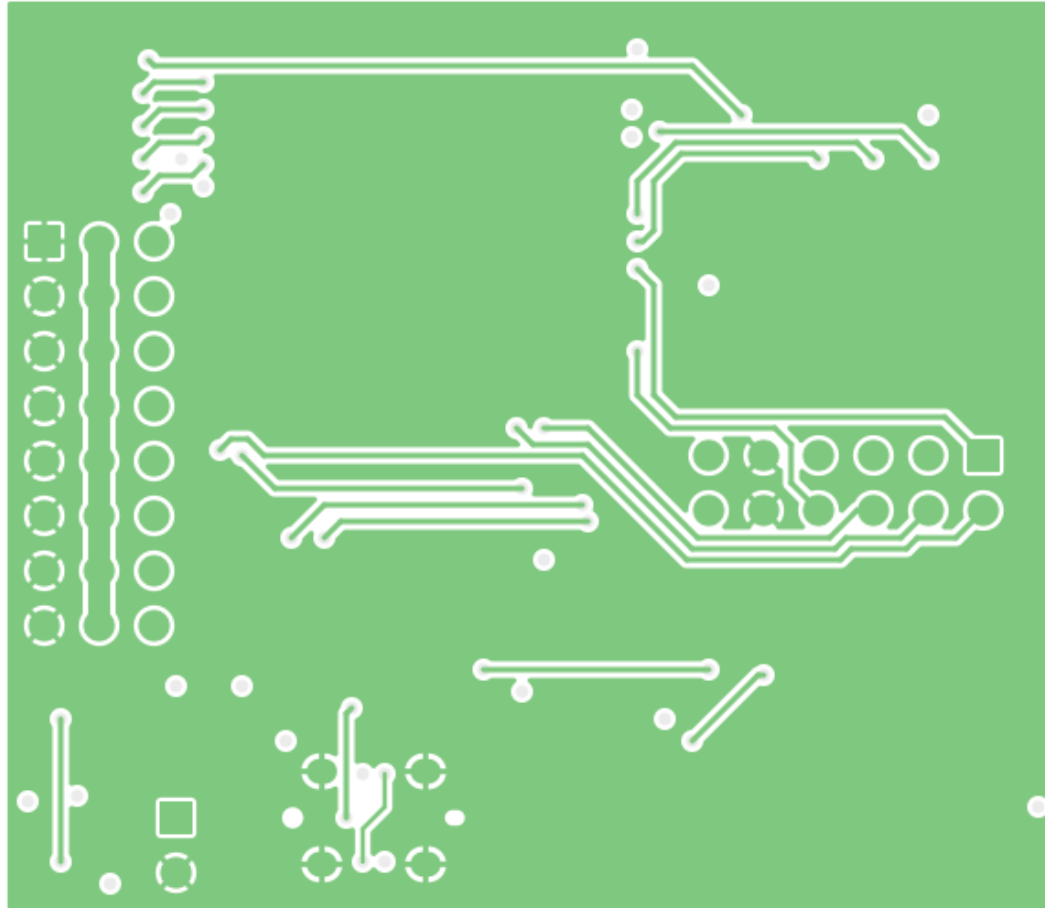
PCBA Extents



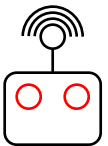
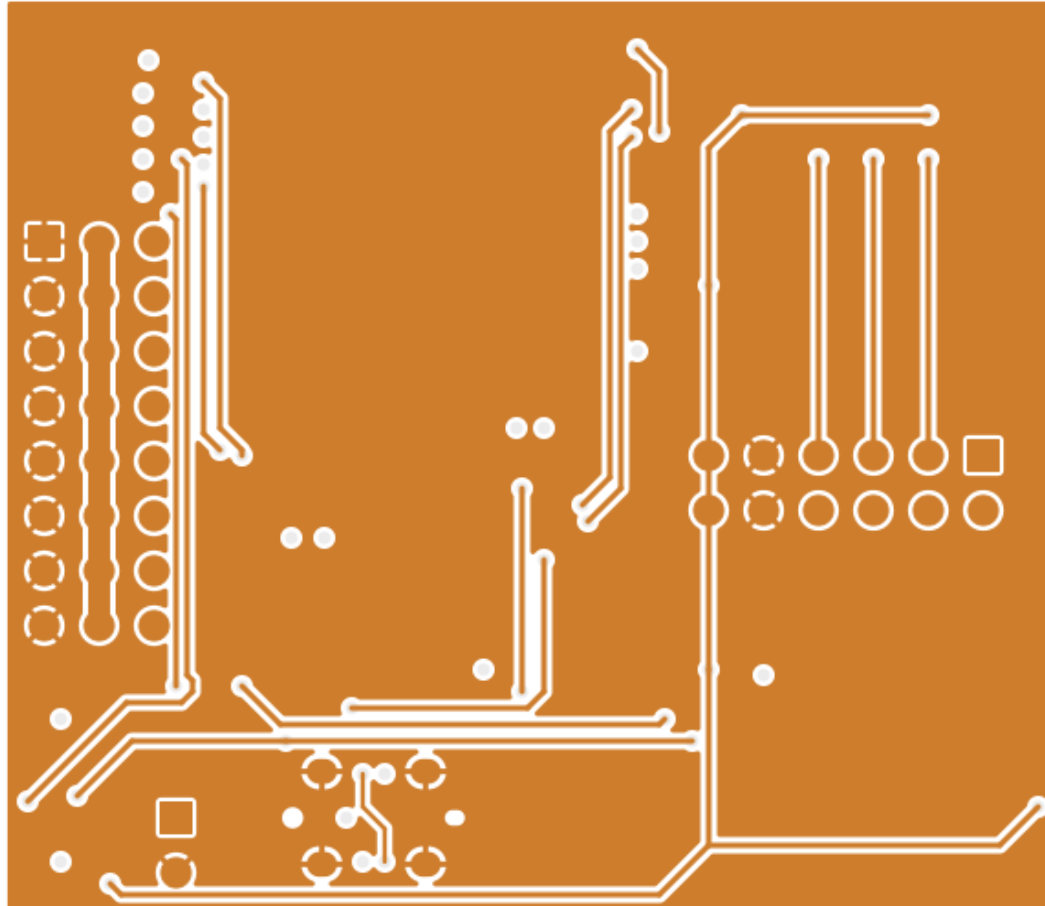
PCBA Top Layer



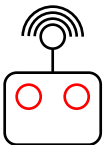
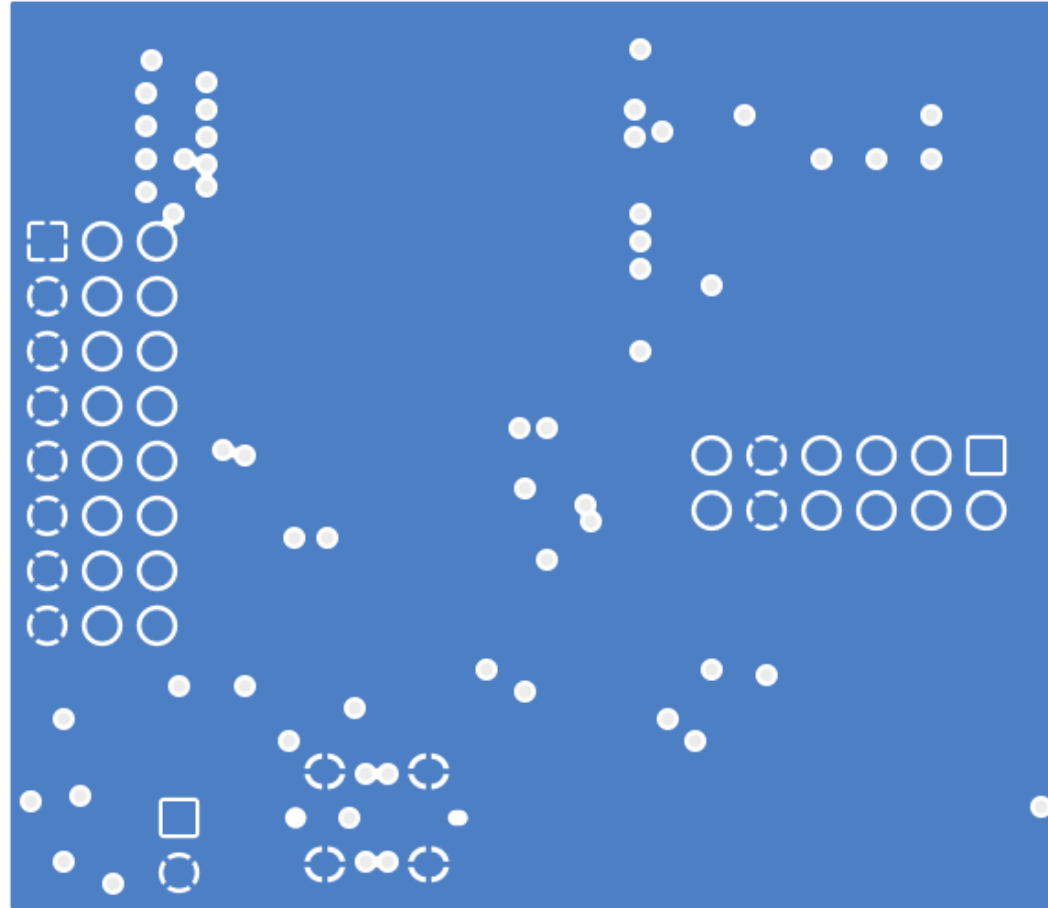
PCBA Layer 1



PCBA Layer 2



PCBA Bottom Layer



WaCS-1 Availability

Developmental units will be made available by contacting the developer Dan Tatum via email dan@wcsrf.com

